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RESEARCH

Development and application of a genetic baseline to inform bull trout (*Salvelinus confluentus*) passage and conservation in the Snake River

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Abstract

Barriers to upstream migration, such as dams, limit fish dispersal in lotic systems, including the federally protected bull trout (*Salvelinus confluentus*). The Hells Canyon Dam on the Snake River currently prevents upstream passage of migratory bull trout into tributary systems in Idaho and Oregon, USA. Viability modeling previously identified the influx of a small number of migrant bull trout as the most important variable for increasing population persistence of resident bull trout populations occupying headwater tributaries above Hells Canyon Dam. Migratory bull trout that overwinter in the Snake River below Hells Canyon Dam could potentially serve as a source of migrants via upstream passage; however, the origin of these fish and their genetic relationship to populations above Hells Canyon Dam is unknown. To address these knowledge gaps, we genotyped bull trout from Idaho and Oregon and developed a genetic baseline to identify the population origination of migratory bull trout below Hells Canyon Dam. An assessment of genetic baseline performance found that stock assignments were highly accurate and minimally biased. Genetic stock identification assigned 97% of the migratory bull trout captured below Hells Canyon Dam to the Imnaha River basin, a large bull trout population located below Hells Canyon Dam. No bull trout were assigned to populations above Hells Canyon Dam. Moving forward, the bull trout genetic data described here can serve as a baseline for future monitoring and conservation efforts in the transboundary waters of Oregon and Idaho and help inform select bull trout passage and translocation efforts.

Keywords *Salvelinus confluentus* · Single nucleotide polymorphisms · Genetic stock identification · Endangered species act · Upstream passage

Introduction

Maintaining connectivity among freshwater habitats is fundamental for fishes, as it allows access to critical habitats, dispersal from natal habitats, and recolonization following disturbances (Selkoe et al. 2016). Artificial structures that reduce or block connectivity (e.g., roads, culverts, and dams) have the potential to reduce population abundance and restrict gene flow (Peterson et al. 2008), which can increase the risk of population extinction due to demographic and environmental stochasticity and genetic processes such as inbreeding and loss of adaptive genetic variation due to random genetic drift (Boyce 1992). A growing body of literature has shown that fish are capable of rapidly recolonizing habitats following barrier removal (e.g., Neville et al. 2016; Fraik et al. 2021) or modification to minimize their effects (Knoth et al. 2022) and that

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movement of fish above barriers (e.g., via human-mediated transport) can maintain gene flow among local populations (DeHaan and Bernall 2013).

The bull trout (*Salvelinus confluentus*) is a freshwater char native to the western United States which can express a migratory life history and is capable of long-distance movements (> 300 km) from lakes or rivers to natal headwater streams where spawning occurs (Fraley and Shepard 1989; Swanberg 1997; Chudnow et al. 2023). A combination of habitat loss and fragmentation, interactions with non-native fish species, and obstruction of migratory corridors have been responsible for significant declines in its abundance relative to historical levels (USFWS 2015). By 1999, all bull trout populations in the coterminous United States were listed as 'threatened' under the US Endangered Species Act (ESA) (64 FR 58910). Habitat fragmentation and the loss of connectivity remain important concerns for bull trout recovery, given the potential for migratory individuals to recolonize extirpated areas and facilitate gene flow among small, isolated populations (USFWS 2015). For example, reduced levels of genetic diversity have been observed in populations of bull trout isolated above barriers (e.g., dams without fish passage; Neraas and Spruell 2001; Whiteley et al. 2006). In addition, bull trout can become entrained in flows through dams, whereby migratory juveniles from habitats upstream of a dam move through or over a dam downstream but are subsequently unable to return upstream as adults to natal tributaries to spawn (Neraas and Spruell 2001; Whiteley et al. 2006; DeHaan et al. 2011a). Efforts to mitigate the effects of fragmentation associated with dams include the construction of fish passage facilities (Noonan et al. 2012), dam removal (Brenkman et al. 2019), and manually transporting fish above dams (i.e., upstream passage; DeHaan and Bernall 2013).

Upstream passage of migratory adults has the potential to maintain distinct life histories within populations of bull trout by minimizing the impact of barriers. For example, approximately a quarter of the juvenile bull trout sampled in two tributaries of the Clark Fork River, Montana, were offspring of adults transported above Cabinet Gorge Dam (DeHaan and Bernall 2013). Critical to the efficacy of upstream passage programs is the ability to assess whether a fish captured below a barrier is of upstream origin. Genetic stock identification uses genotype frequencies from a suite of potential source populations to infer the source of origin of an unknown individual (Anderson et al. 2008) and is particularly well suited for species that display strong population genetic structure, such as bull trout (Costello et al. 2003; Whiteley et al. 2006; Ardren et al. 2011). To date, genetic stock identification has been successfully applied to bull trout to address a variety of management- and research-related questions (e.g., DeHaan et al. 2011a; Warnock et al. 2011; DeHaan and Bernall 2013).

The Hells Canyon Dam on the Snake River along the Idaho-Oregon border was completed in 1967 and is owned and operated by the Idaho Power Company. Bull trout occupy headwater streams above Hells Canyon Dam, including those in the Pine Creek, Indian Creek, and Wildhorse River basins. Bull trout populations above Hells Canyon Dam are generally dominated by resident fish that are smaller in size and have lower fecundity relative to migratory individuals (Pratt 2003). As a result of isolation, low abundance, and limited suitable habitat, bull trout populations in Pine and Indian Creek are at risk of extinction. Population modeling identified the annual immigration of small numbers of mature migratory females (1–6) can significantly increase the likelihood of persistence (Pratt 2003). Migratory bull trout are seasonally present in the Snake River below Hells Canyon Dam and could potentially serve as a source of migrants for upstream populations if habitat above the dam were accessible. However, migratory bull trout cannot pass upstream of Hells Canyon Dam. With the issuance of a new federal operation license expected in 2025–2026, Idaho Power Company will improve existing salmon and steelhead trap-and-haul facilities at Hells Canyon Dam, enabling upstream passage of migratory bull trout into Hells Canyon Reservoir, into which the Pine Creek and Indian Creek basins flow (Fig. 1). Oxbow Dam is located upstream of Hells Canyon Reservoir, and the Wildhorse River terminates at Oxbow Reservoir. Passage into the Wildhorse River could occur after bull trout passage objectives have been achieved in the Pine and Indian Creek basins and facilities are constructed at Oxbow Dam. In combination, there are thus several current and future opportunities to minimize the impacts of barriers in the Hells Canyon reach of the Snake River and to promote bull trout conservation through the maintenance of multiple life histories (resident and migratory).

Previous work has described the population genetic characteristics of bull trout above (Pine and Indian Creeks; Spruell et al. 2003) and below Hells Canyon Dam (e.g., Grande Ronde and Imnaha river basins; Spruell et al. 2003; Ardren et al. 2011; Hudson et al. 2017; Whitesel et al. 2022), but several questions relevant to upstream translocation of bull trout in Hells Canyon remain unanswered. For instance, the source and origin of migratory bull trout below Hells Canyon Dam are unknown, as is their genetic relatedness to upstream populations. Additionally, greater detail of the genetic relationships among, and levels of diversity within, bull trout populations in the Hells Canyon reach of the Snake River could help identify at-risk populations and serve as a baseline for monitoring future responses to upstream passage efforts.

Our study objective was to develop a bull trout population genetic baseline from core areas above and below the Hells Canyon Complex dams, describe levels of genetic diversity

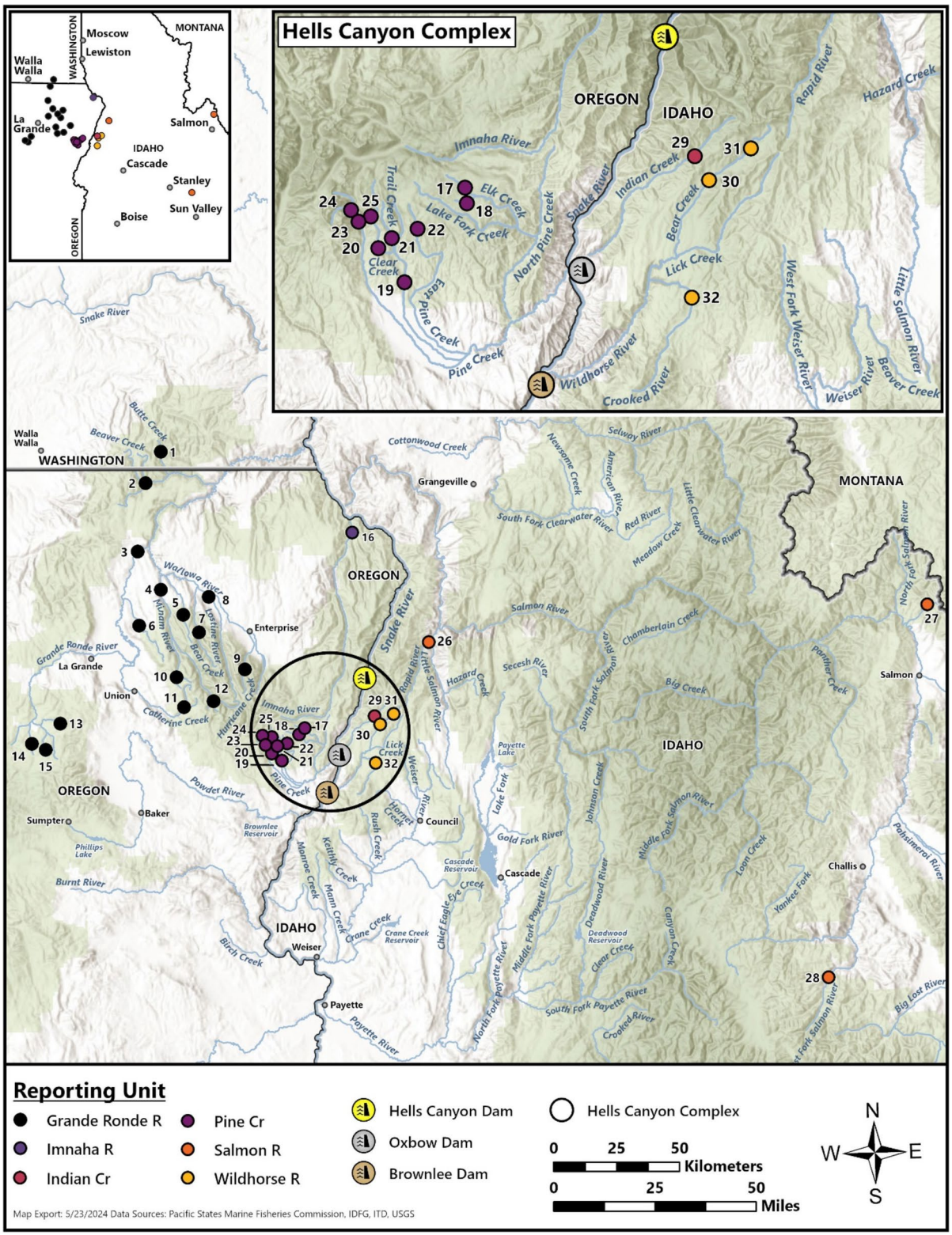

Fig. 1 Populations of bull trout (*Salvelinus confluentus*) sampled to construct a genetic baseline for use in the Snake River basin. Numbers correspond to populations described in Table 1

and differentiation within and among populations, and assess the utility of this baseline for genetic stock identification purposes. Using this newly constructed baseline, we inferred the genetic stock of migratory bull trout collected from the Snake River below Hells Canyon Dam. These genetic resources may directly apply to current and future conservation efforts for bull trout above Hells Canyon Dam, including upstream passage and translocations.

Methods

Study area description

We focused our study on the mainstem Snake River along the Idaho, Oregon, and Washington borders and on tributaries to the Snake River that originate primarily in northeast Oregon and central and west-central Idaho. Downstream of Hells Canyon Dam, the study area included the Grande Ronde, Imnaha, and Salmon River basins and a 127-km reach of the mainstem Snake River from Hells Canyon Dam to the confluence of the Grande Ronde River. Upstream of Hells Canyon Dam, the study area encompassed the Pine Creek, Indian Creek, and Wildhorse River basins. Pine and Indian creeks are tributaries to Hells Canyon Reservoir, and the Wildhorse River flows into Oxbow Reservoir. The Grande Ronde, Imnaha, and Salmon rivers are undammed tributaries to the Snake River with relatively large drainage areas (approximately 200,000–3,600,00 ha) supporting resident and migratory bull trout populations. Pine Creek, Indian Creek, and the Wildhorse River are comparatively small tributaries (< 100,000 ha) that contain only resident bull trout populations in their headwaters.

The U.S. Fish and Wildlife Service (USFWS) recovery plan for bull trout in the coterminous United States (USFWS 2015) includes the population from the Hells Canyon portion of the Snake River as part of the Mid-Columbia Recovery unit, which encompasses 24 core areas and spans four geographic regions. Two core areas were designated in the vicinity of the three-dam Hells Canyon Hydroelectric Complex (HCC; Hells Canyon, Oxbow, and Brownlee dams), including (1) Powder River basin, and (2) Pine Creek, Indian Creek, and Wildhorse River basins. Historically, the Pine, Indian, and Wildhorse basins likely supported multiple life histories (resident and migratory forms); however, resident fish currently dominate most local populations (Buchanan et al. 1997; Chandler et al. 2001, 2003; Wilkison and Trainer 2017; IPC 2020). Bull trout fitted with radio tags in Hells Canyon Reservoir were detected moving into Pine Creek which suggests some remnant migratory life history may persist in the Indian Creek or Pine Creek basins (Chandler et al. 2003; IPC 2020). The abundance of bull trout in Pine and Indian Creeks is low. As noted previously, demographic modeling has indicated that adding a few migratory individuals (as low as 1–6) can significantly increase their likelihood of persistence (ODFW 2005; Pratt 2003).

A population of migratory bull trout is seasonally present in the Snake River below Hells Canyon Dam. These fish could have been born in the Salmon River basin (Rapid River), the Imnaha River basin, or the Grande Ronde River basin. The seasonal movements of migratory bull trout from below Hells Canyon Dam have been tracked via radio telemetry and indicated residence in the mainstem Snake River below Hells Canyon Dam during the fall and winter (e.g., October through February), migration into tributary river systems from mid-March to late-May, and arrival at headwater reaches (i.e., spawning grounds) June through August (Chandler et al. 2001, IPC 2020). Over half (57%) of the 47 migratory bull trout tagged below Hells Canyon Dam entered the Imnaha River during the summer months (IPC 2020), suggesting this is a potential source population. As noted, the Hells Canyon Dam lacks upstream passage facilities for bull trout. Still, facilities to trap and haul migratory fish above the dam will be operational after issuing a new federal operating license.

Field sampling

Bull trout were sampled in wadable streams via backpack electrofishing by biologists and personnel from the US Forest Service, Idaho Power Company, Idaho Department of Fish and Game, and Oregon Department of Fish and Wildlife. Fish were anesthetized before being sampled for a ~ 1 cm^2 fin clip for genetic analysis. Genetic samples were preserved in 95% ethanol or on Whatman sheets (Fisher Scientific, Catalog No. 05-714-4; LaHood et al. 2008). All fish were released alive.

In addition to bull trout collected for constructing a genetic baseline, samples of migratory bull trout in the Snake River below Hells Canyon Dam were collected by staff from Idaho Power Company. Samples were collected during the overwintering period, when migratory bull trout reside in the mainstem river, by hook-and-line angling using lures with barbless hooks. Genetic tissues were collected following the same procedures used for the genetic baseline. Migratory bull trout from the Snake River were collected independently of efforts to collect samples for our genetic baseline and were assumed to have originated from one or more potential source populations in our baseline.

Laboratory procedures

Genomic DNA was extracted using the nexttec Genomic DNA Isolation Kit (XpressBio, Thurmont, Maryland) following the manufacturer's protocol. A panel of 266 single nucleotide polymorphism (SNP) loci described in Bohling

et al. (2021) were amplified using the genotyping-in-thousands by sequencing (GT-seq) protocol (Campbell et al. 2015). Of the 266 loci, one is a male-specific sequence, two are mitochondrial loci, 20 are nuclear loci diagnostic between brook trout (*S. fontinalis*) and bull trout, eight are loci selected for allele frequency differences between *Salvelinus* spp.. The remaining 235 are autosomal, nuclear loci containing SNPs developed for range-wide bull trout monitoring. Population genetic analyses described below were performed using the putatively neutral autosomal SNPs. To ensure that only bull trout were included in analyses, samples were required to have genotypes for a minimum of 10 of the 20 brook trout × bull trout diagnostic SNPs, and samples with more than one brook trout allele were excluded. After removing brook trout and hybrids, we retained bull trout that were successfully genotyped at ≥ 90% of loci.

Genetic diversity and differentiation

Bull trout were grouped into collections based on geographic location; each was intended to represent a random mating population (Table 1; Fig. 1). In instances where the same location was sampled multiple times, we pooled individuals across sampling events, which was supported by low metrics of genetic differentiation (data not shown).

Table 1 Details for the Snake River bull trout (*Salvelinus confluentus*) genetic baseline organized by reporting unit

Reporting unit	Map no	Population	Years sampled	n	H_E	H_O	F_{IS}	P
Grande Ronde R	1	Butte Cr	1995	16	0.110	0.113	– 0.034 (– 0.106, – 0.017)	0.362
	2	Wenaha R	1995, 2003, 2004, 2020	100	0.137	0.132	0.018 (– 0.005, 0.033)	0.549
	3	Lookingglass Cr	2003	26	0.111	0.107	0.030 (– 0.036, 0.053)	0.417
	4	Minam R	2018–2020	38	0.134	0.125	0.066 (0.011, 0.090)	0.502
	5	Deer Cr	2003	16	0.077	0.082	– 0.050 (– 0.151, – 0.033)	0.251
	6	Indian Cr	1995	20	0.093	0.097	– 0.041 (– 0.099, – 0.020)	0.340
	7	Bear Cr	1995	19	0.111	0.118	– 0.037 (– 0.134, – 0.012)	0.387
	8	Lostine R	1995, 2003, 2018–2020	386	0.120	0.117	0.064 (0.026, 0.077)	0.562
	9	Hurricane Cr	1995, 2003	46	0.065	0.065	– 0.008 (– 0.061, 0.024)	0.174
	10	Little Minam R	1995, 2003	54	0.089	0.089	– 0.001 (– 0.034, 0.020)	0.315
	11	North Fork Catherine Cr	1995	26	0.111	0.113	– 0.014 (– 0.066, 0.002)	0.336
	12	Elk Cr	1995	20	0.090	0.090	– 0.002 (– 0.099, 0.038)	0.298
	13	Limber Jim Cr	1995, 2003	20	0.043	0.048	– 0.081 (– 0.168, – 0.045)	0.183
	14	Indiana Cr	2003	26	0.044	0.042	0.042 (– 0.057, 0.083)	0.179
	15	Clear Cr	1995, 2003	52	0.079	0.082	– 0.008 (– 0.053, 0.009)	0.306
Imnaha R	16	Imnaha R	2011, 2012	54	0.138	0.136	0.006 (– 0.023, 0.022)	0.545
Pine Cr	17	Elk Cr	2015	60	0.102	0.101	0.013 (– 0.022, 0.030)	0.383
	18	Aspen Cr	2015	32	0.092	0.091	0.004 (– 0.056, 0.037)	0.302
	19	Clear Cr	2013	65	0.112	0.111	– 0.006 (– 0.039, 0.016)	0.362
	20	Meadow Cr	2016	54	0.108	0.100	0.090 (0.032, 0.115)	0.319
	21	Trail Cr	2016	66	0.067	0.066	0.008 (– 0.033, 0.039)	0.230
	22	East Pine Cr	2016	130	0.059	0.054	0.076 (0.049, 0.096)	0.336
	23	Pine Cr	2014	58	0.067	0.065	0.045 (0.000, 0.065)	0.289
	24	Middle Fork Pine Cr	2014	24	0.079	0.074	0.059 (– 0.018, 0.090)	0.238
	25	East Fork Pine Cr	2014	180	0.030	0.031	0.020 (– 0.033, 0.041)	0.234
Indian Cr	29	Indian Cr	2008, 2011, 2019	68	0.052	0.051	0.014 (– 0.033, 0.046)	0.183
Wildhorse R	30	Bear Cr	2017	12	0.036	0.035	0.050 (– 0.103, 0.104)	0.098
	31	Wesley Cr	2017	54	0.037	0.036	– 0.013 (– 0.074, 0.033)	0.111
	32	Crooked R	2018	50	0.071	0.072	– 0.025 (– 0.068, 0.000)	0.234
Salmon R	26	Rapid R	2004	59	0.144	0.138	0.019 (– 0.005, 0.033)	0.519
	27	4th of July Cr	2004	30	0.121	0.120	0.017 (– 0.037, 0.033)	0.434
	28	East Fork Salmon R	2008–2014	977	0.121	0.116	0.040 (0.027, 0.045)	0.451

For each population the number of samples (n), expected heterozygosity (H_E), observed heterozygosity (H_O), inbreeding coefficient (F_{IS}) with 95% confidence intervals in parentheses, and proportion of markers that were polymorphic (P) per population are provided. Map number (Map no.) refers to locations presented in Fig. 1

Exact tests were performed to determine whether genotype frequencies conformed to Hardy–Weinberg equilibrium (HWE) at the population level using the hw.test in the R package (R Core Team 2023) pegas (Paradis 2010). *P*-values associated with HWE tests were corrected for multiple comparisons using the false discovery rate approach of Benjamini and Hochberg (1995). Genetic diversity was assessed at the population and reporting unit levels using expected heterozygosity (H_E) and observed heterozygosity (H_O) of all 235 SNPs regardless of whether they were polymorphic in the given population. The proportion of polymorphic loci (*P*) was calculated as a measure of genetic variation within the population. Inbreeding was assessed by calculating F_{IS} utilizing the R package diveRsity v1.9.90 (Keenan et al. 2013), and ninety-five percent confidence intervals for F_{IS} were estimated via 999 bootstrap replicates.

Levels of genetic differentiation were quantified at the reporting unit and population levels as F_{ST} using the method of Weir and Cockerham (1984) as implemented in hierfstat (Goudet 2005). For reporting unit-level estimates of F_{ST}, all populations were collapsed within a reporting unit before estimating differentiation.

Genetic relationships among bull trout reporting units were visualized using discriminant analysis of principal components (DAPC), which first reduces the dimensionality of the data via a principal components analysis and then applies discriminant functions (DFs) to the transformed data to minimize within-group variance and maximize between-group variance (Jombart et al. 2010). Because DAPC can be sensitive to the number of retained principal components (PCs), we performed stratified cross-validation with 1000 replicates for 10–110 retained PCs using the *xvalDAPC* function from adegenet (Jombart 2008). We retained the number of PCs that resulted in the lowest root-mean square error during cross-validation. Reporting units were used as a priori clusters for DAPC, and all discriminant functions were retained.

Genetic baseline performance and stock identification

Genetic stock identification (GSI) was performed by calculating mixture proportions and individual assignments were inferred using the Bayesian method described by Moran and Anderson (2019). First, self-assignment tests were performed using the leave-one-out over multilocus genotypes method of Anderson et al. (2008) to assess baseline performance. Each fish in the baseline was sequentially removed, allele frequencies were recalculated, and the most likely origin of the removed fish was then estimated. As part of this procedure, each individual was assigned to a reporting unit and population within a reporting unit. Next, the accuracy of individual assignments was evaluated via 200 mixture simulations, in which mixtures of 50 individuals of known, true origin were generated, and GSI was used to estimate the origin of these 10,000 simulated individuals. Fish were assigned to the reporting unit and population with the highest posterior probability (i.e., no minimum threshold was used for retaining an assignment). To evaluate the accuracy of inferred mixture proportions, we performed 500 simulations of mixtures containing 200 individuals and compared the true and estimated mixture proportions. For all simulations, true mixture proportions were chosen by taking random samples from a Dirichlet distribution with all parameters equal to 1.5 first to determine the true proportions of reporting units and then select the true proportions of populations within each reporting unit.

Following the evaluation of baseline accuracy, GSI was performed on a set of bull trout multilocus genotypes from the Snake River below Hells Canyon Dam that were collected independently of our baseline. Individual assignments were generated based on 2000 burn-ins and 10,000 sampled MCMC iterations. Individuals were assigned to the population and reporting unit with the highest posterior probabilities. Genetic stock assignments and simulations were performed using the R package *rubias* (Moran and Anderson 2019).

Unless stated otherwise, statistical analyses were performed using the statistical package R (R Core Team 2023).

Results

Genotyping

A total of 3819 fin clips were genotyped, which included both bull trout used to construct a genetic baseline and samples of unknown origin (presumptive mixture samples from the Snake River below Hells Canyon Dam). Of these samples, 149 failed to amplify at the minimum number of hybrid markers, 324 were identified as brook trout, and 91 were identified as hybrids. Following hybrid screening, 3255 samples were identified as bull trout, and 87 were removed from the analysis due to insufficient genotyping completeness (i.e., failed to amplify at ≥ 90% of loci). Of the 3168 samples that passed all filtering steps, 2838 were used to construct the GSI baseline. The remaining 330 samples were from fish collected from the Snake River below Hells Canyon Dam and were assigned to genetic stocks via GSI.

Genetic diversity and differentiation

Following false discovery correction, we observed 88 deviations from HWE, representing 1.2% of the 7520 locus × population combinations. Deviations from HWE occurred at one or more loci in 22 of the 32 populations. Among populations

that deviated from HWE, the Lostine River population had the highest number of deviations (n = 19 loci, 8.1% of all loci), while all other populations deviated at 10 or fewer loci. A total of 52 loci deviated from HWE, but only two loci (SACO_RAD_20371 and Sco_RAD_12481) deviated in more than five populations. Forty-two of the 52 loci deviated from HWE in a single population.

Levels of genetic diversity varied across reporting units and were highest in the Imnaha River (H_O = 0.136), followed by the Salmon River (H_O = 0.125) and Grande Ronde River (H_O = 0.095; Table 1; Fig. 2). Reporting units from headwater tributaries above Hells Canyon Dam displayed generally lower levels of mean observed heterozygosity (Pine Cr, H_O = 0.077; Indian Cr, H_O = 0.051; Wildhorse River, H_O = 0.048) than downstream reporting units. Samples were collected from four headwater tributaries within the Pine Creek reporting unit (Pine Creek, Clear Creek, East Pine Creek, and Elk Creek; Fig. 1), and levels of observed heterozygosity varied within and among these tributaries. For example, East Fork Pine Creek (a tributary to Pine Creek) displayed the lowest level of observed heterozygosity of any population (H_O = 0.031). In contrast, other collections from different tributaries within Pine Creek were similar to mean levels of diversity across all reporting units (Middle Fork Pine Creek, H_O = 0.079; mean H_O across all collections = 0.088). Other reporting units also contained populations with high and low genetic diversity levels. The extent to which heterozygosity estimates varied among populations was lowest in the Salmon River (H_O range = 0.116–0.138) and Wildhorse River (H_O range = 0.035–0.072) and highest in the Grande Ronde River (H_O range = 0.042–0.132) and in Pine Creek (H_O range = 0.030–0.112; Fig. 2). Patterns in the proportion of polymorphic (P) loci per population followed those of observed heterozygosity. Specifically, we observed variation among populations within reporting units (Table 1), and the proportion of loci that were polymorphic averaged across populations was highest for the Imnaha River (0.544), Salmon River (0.468), and Grande Ronde River (0.344) reporting units. In contrast, the proportion of polymorphic loci was lower in reporting units above Hells Canyon Dam (average P across populations: Pine Creek = 0.299, Indian Creek = 0.183, Wildhorse River = 0.148).

Levels of inbreeding (F_{IS}) calculated at the population level were generally close to 0, suggesting that populations were breeding approximately randomly (Table 1).

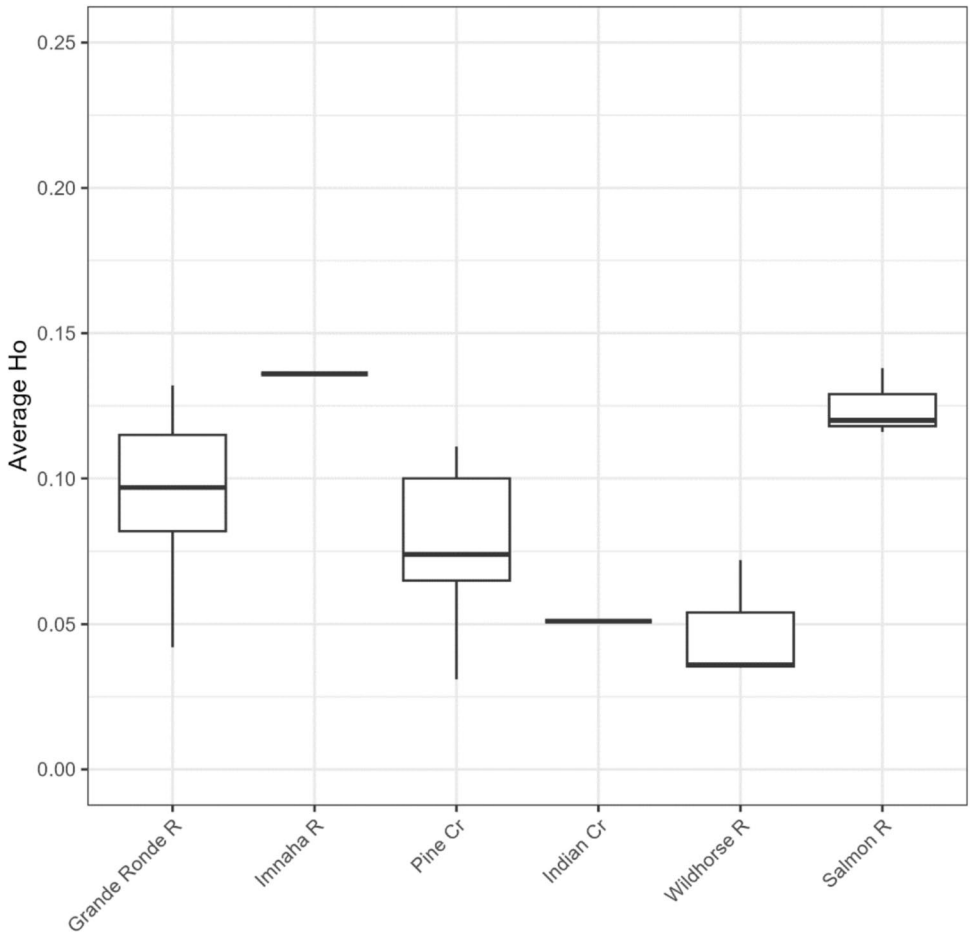

Fig. 2 Levels of observed heterozygosity (H_O) for populations of bull trout (*Salvelinus confluentus*) sampled from the Snake River basin organized by reporting unit. Solid horizontal bars denote median values, and the lower and upper hinges correspond to the first and third quartiles (the 25th and 75th percentiles)

Estimated confidence intervals were not significantly different from zero for 22 of the 32 populations. Among the ten populations with F_{IS} that deviated significantly from zero, five estimates of F_{IS} were positive (heterozygote deficit), and five were negative (heterozygote excess). Mean levels of inbreeding at the reporting unit level were lowest for the Grande Ronde River (F_{IS} = -0.012), followed by Indian Creek (F_{IS} = 0.009), Wildhorse River (F_{IS} = 0.010), Imnaha River (F_{IS} = 0.015), Pine Creek (F_{IS} = 0.025) and the Salmon River (F_{IS} = 0.031).

We observed evidence of strong genetic structuring among bull trout reporting units with an average pairwise F_{ST} of 0.267 (range = 0.088–0.469; Fig. 3). The Salmon River reporting unit was the most strongly differentiated (average pairwise F_{ST} = 0.416, range = 0.363–0.469), whereas the Grande Ronde River was the least differentiated (average pairwise F_{ST} = 0.209, range = 0.088–0.363). Above Hells Canyon Dam, Pine Creek had the lowest average pairwise F_{ST} (0.226), followed by the Wildhorse River (0.248) and Indian Creek (0.283). Pine Creek was most closely related to the Imnaha River (pairwise F_{ST} = 0.142), Indian Creek was most similar to the Wildhorse River (pairwise F_{ST} = 0.215), and the Wildhorse River was the least differentiated from Pine Creek (pairwise F_{ST} = 0.171).

At the population level, we observed considerable variation in levels of pairwise genetic differentiation. For reporting units with samples from multiple populations, we documented weakly and strongly differentiated populations within the same reporting unit (Fig. 4). For example, Limber Jim, Indiana, and Elk creeks within the Grande Ronde River reporting unit had large average pairwise F_{ST} values of ~ 0.5, which were nearly double those of the Lostine (F_{ST} = 0.247), Minam (F_{ST} = 0.246) and Wenaha rivers (F_{ST} = 0.247). Populations above Hells Canyon Dam had an average pairwise F_{ST} of 0.384, slightly larger than populations below Hells Canyon Dam (0.354).

Based on cross-validation, the optimum number of PCs retained for DAPC was 65, which conserved 88.45% of the observed variance. The majority of variance in the discriminant analysis was explained by DFs 1 and 2 (DF 1 = 62.35%, DF 2 = 25.85%). Discriminant function 1 separated the Salmon River from all other reporting units, and DF 2 distinguished the Grande Ronde and Imnaha Rivers from Indian Creek, Pine Creek, and Wildhorse River (Fig. 5a). Most individuals from the Salmon River clustered tightly together. Still, dots representing individuals from the Salmon River reporting unit sampled in Rapid River were observed to be located between the Salmon River and Imnaha River

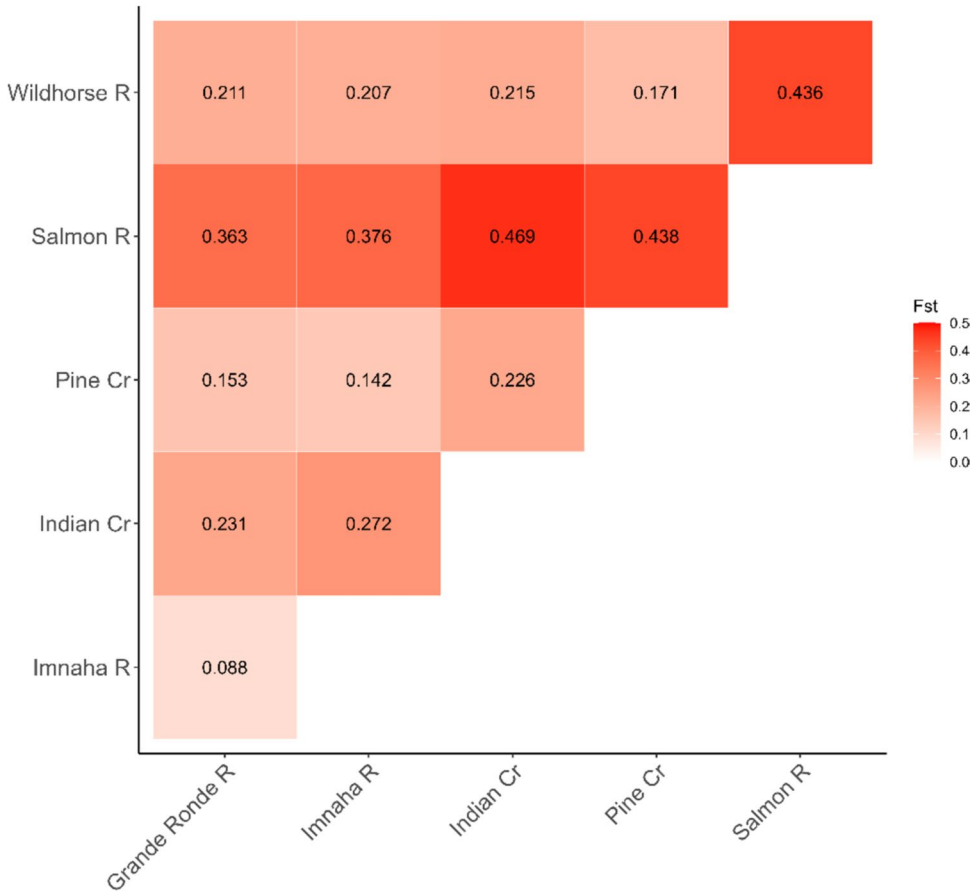

Fig. 3 Average pairwise genetic differentiation (F_{ST}) between bull trout (*Salvelinus confluentus*) populations sampled from tributary systems above Hells Canyon Dam (Pine Cr, Indian Cr, Wildhorse R) as well as downstream populations in the Grande Ronde, Imnaha, and Salmon rivers

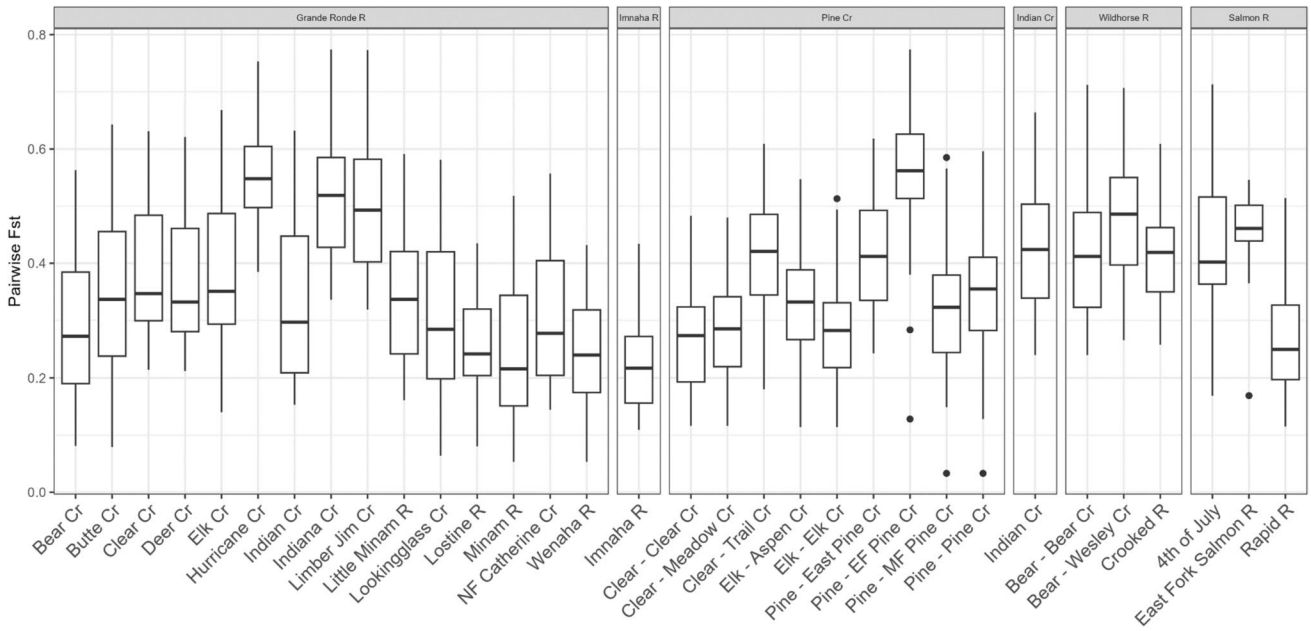

Fig. 4 Levels of pairwise genetic differentiation (F_{ST}) at the population level for bull trout (*Salvelinus confluentus*) sampled from tributary systems above Hell Canyon Dam (Pine Cr, Indian Cr, Wildhorse R) as well as downstream populations in the Grande Ronde, Imnaha, and Salmon Rivers. Solid horizontal bars denote median values, and the lower and upper hinges correspond to the first and third quartiles (the 25th and 75th percentiles)

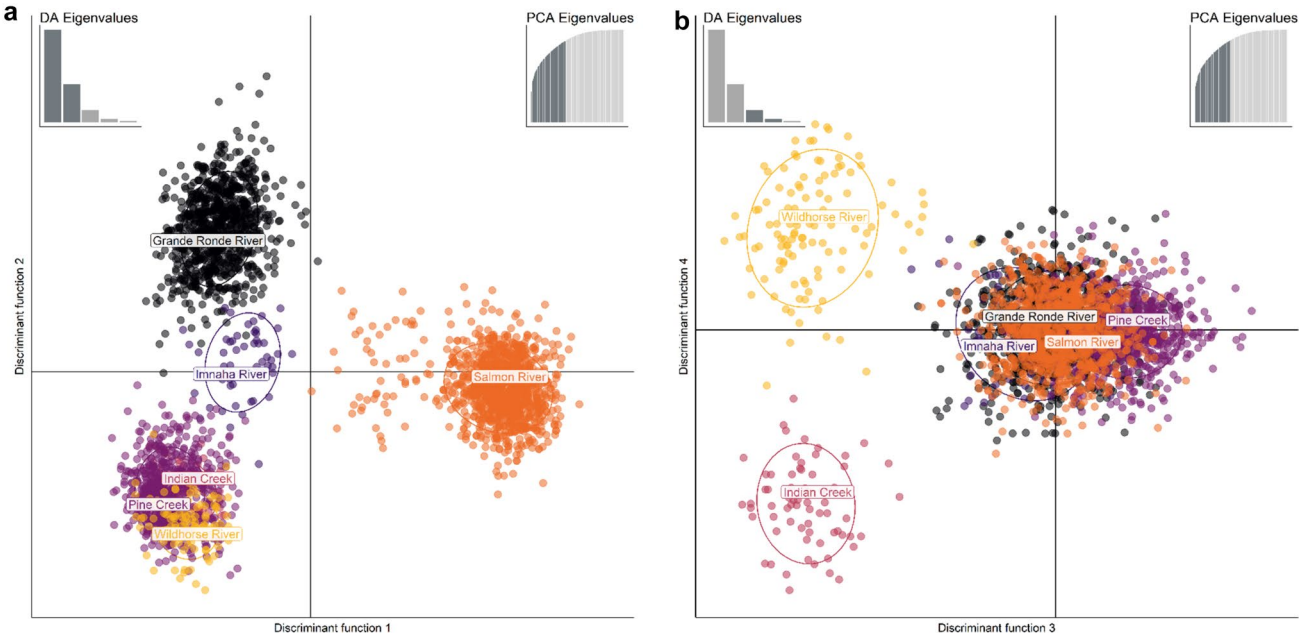

Fig. 5 Plots of discriminant analysis of principal components (DAPC) using reporting units as a priori genetic clusters of bull trout (*Salvelinus confluentus*) sampled from the Snake River basin. Each point represents an individual, and colors of points and inertia ellipses correspond to bull trout reporting units. **a** DAPC plotted on axes of discriminant function 1 and discriminant function 2. **b** DAPC plotted on axes of discriminant function 3 and discriminant function 4

clusters, likely due to geographic proximity between Rapid River and the Imnaha River. While individuals from the Grande Ronde River and Imnaha River formed relatively discrete clusters, there was a high degree of overlap among the Indian Creek, Pine Creek, and Wildhorse River clusters along axes of DFs 1 and 2, indicating lower levels of

differentiation among reporting units above Hells Canyon Dam. However, dots representing many individuals from the Pine Creek reporting unit occupied space intermediate between the Pine Creek and Imnaha River clusters, indicating higher genetic similarity among some individuals within these two clusters. Clusters for Indian Creek, Pine Creek, and Wildhorse River were separated along axes of DFs 3 and 4 (Fig. 5b), though these DFs explained less variance. Overall, relationships among reporting units were inferred from DAPC-aligned locations of populations in geographic space and with patterns of genetic structure observed via comparisons of heterozygosity and pairwise F_{ST}.

Estimated accuracy of GSI

All but three individuals (<0.1% of 2838 individuals) in our genetic baseline were self-assigned to their reporting unit of origin (Table 2). Two individuals sampled in the Grande Ronde River were assigned to the Imnaha River reporting unit, and 1 individual from the Imnaha River was assigned to the Grande Ronde River. At the population level, self-assignment rates were greater than 90% for 29 of the 32 collections. Two populations in the Pine Creek reporting unit (Pine Cr and MF Pine Cr) had low self-assignment rates (41.4 and 37.5%, respectively); however, all fish that did not self-assign to their population of origin were assigned to other populations within the same reporting unit. Across all populations, the self-assignment rate at the population level was 96.5%.

The accuracy of individual assignments to reporting units was almost perfect, with only five (out of 10,000) simulated individuals mis-assigning (Table 3). Inferred mixture proportions at the population level appeared unbiased, except for the Minam River, which was biased low (results not shown). Estimates of mixture proportions for reporting units were unbiased (Fig. 6).

GSI assignments

Of the 286 migratory bull trout samples collected from the Snake River, 279 (97.6%) were assigned to the Imnaha River basin via GSI. One fish (0.3%) was assigned to the Grande Ronde River reporting unit (Wenaha River), and six assignments (2.1%) were made to the Salmon River basin (Rapid River). No migratory samples from the Snake River were assigned to reporting units above Hells Canyon Dam (Pine Creek, Indian Creek, or Wildhorse River) (Table 4).

Discussion

Our genetic baseline revealed that levels of genetic diversity within bull trout populations varied considerably among reporting units. Populations above Hells Canyon Dam displayed lower levels of diversity than downstream reporting

Table 2 The number of bull trout (*Salvelinus confluentus*) that were self-assigned to reporting units (with percentages in parentheses) using the leave-one-out approach implemented in *rubias* (Moran and Anderson 2019)

	Inferred reporting group					
Known reporting group	Grande Ronde R	Imnaha R	Pine Cr	Indian Cr	Wildhorse R	Salmon R
Grande Ronde R	863 (0.99)	2 (0.01)	–	–	–	–
Imnaha R	1 (0.02)	53 (0.98)	–	–	–	–
Pine Cr	–	–	68 (1.00)	–	–	–
Indian Cr	–	–	–	669 (1.00)	–	–
Wildhorse R	–	–	–	–	1066 (1.00)	–
Salmon R	–	–	–	–	–	116 (1.00)

Table 3 The number of individual bull trout (*Salvelinus confluentus*) and percentages of individuals correctly assigned (in parentheses) to different reporting units based on simulated mixtures of individuals (n = 10,000 individuals) using the leave-one-out approach implemented in *rubias* (Moran and Anderson 2019)

	Inferred reporting group					
Known reporting group	Grande Ronde R	Imnaha R	Pine Cr	Indian Cr	Wildhorse R	Salmon R
Grande Ronde R	1744 (99.9)	2 (0.1)	–	–	–	–
Imnaha R	2 (0.1)	1782 (99.9)	–	–	–	–
Pine Cr	–	–	1523 (100)	–	–	–
Indian Cr	–	–	–	1624 (99.9)	1 (0.1)	–
Wildhorse R	–	–	–	–	1523 (100)	–
Salmon R	–	–	–	–	–	1799 (100)

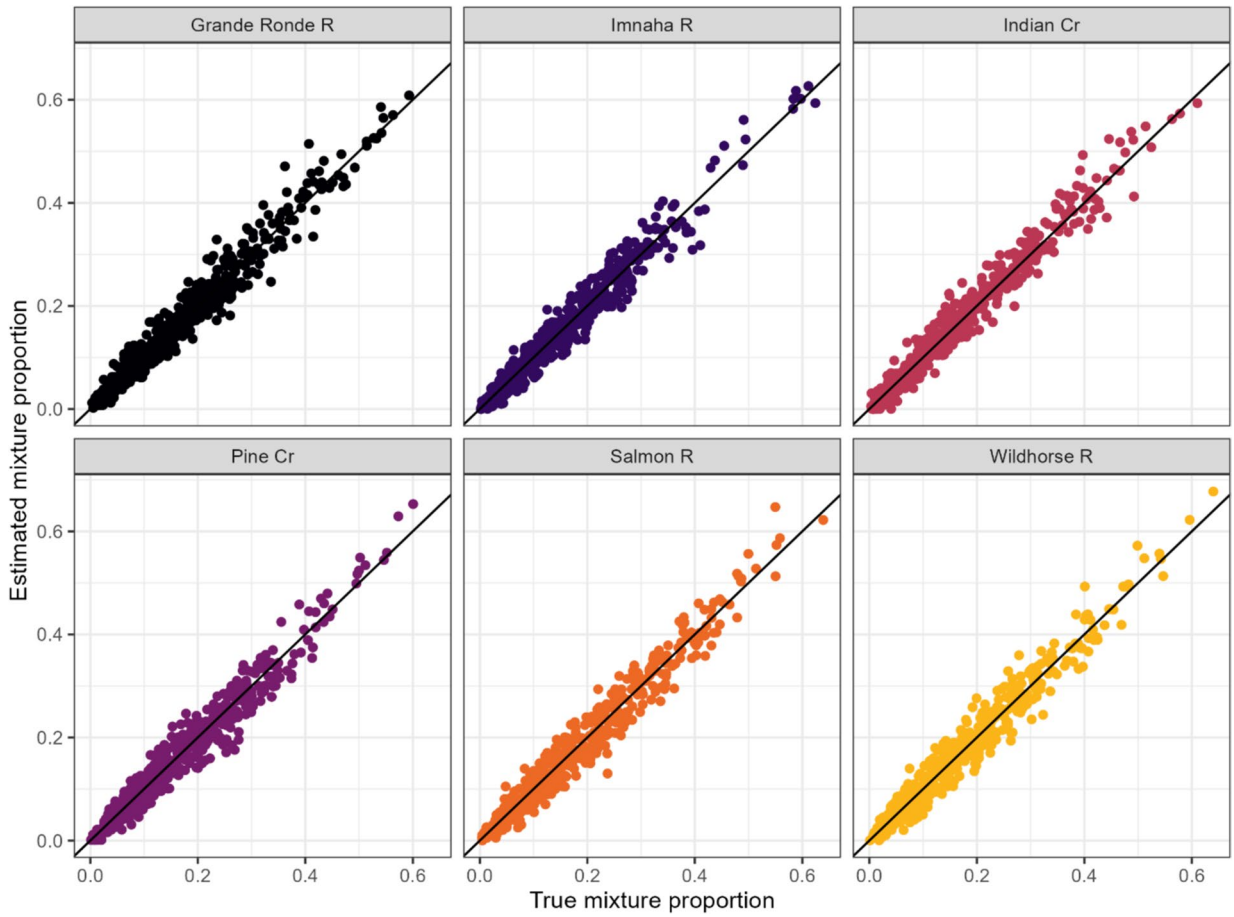

Fig. 6 A comparison of true mixture proportions and simulated mixture proportions by reporting unit for populations of bull trout (*Salvelinus confluentus*) sampled from the Snake River and its tributaries

Table 4 Reporting unit of origin for samples collected below Hells Canyon Dam based on genetic stock identification

Collection year	n	Assigned reporting unit		
		Grande Ronde R	Imnaha R	Salmon River
2001	13	0	12	1
2006	12	0	12	0
2007	21	0	21	0
2008	41	0	41	0
2009	39	0	39	0
2010	23	0	23	0
2011	28	0	28	0
2012	56	1	54	1
2015	33	0	32	1
2016	17	0	14	3
2017	3	0	3	0

units, which a combination of population size and gene flow could explain. Pine Creek, Indian Creek, and Wildhorse River basins encompass smaller amounts of suitable habitat relative to other reporting units in our study (e.g., average spawning distribution for populations in Pine Creek ~ 13 km, Grande Ronde River ~ 32 km, Imnaha River ~ 75 km; ODFW 2005). Levels of genetic diversity in bull trout populations are positively correlated with patch size at both small (e.g., Glacier National Park; Meeuwig et al. 2010) and broad geographic scales (Columbia River basin; Kovach et al. 2015), although exceptions have been noted (Costello et al. 2003; Whiteley et al. 2006). Therefore, the limited habitat available in the Pine Creek, Indian Creek, and Wildhorse River basins may support smaller populations, which are more likely to experience losses in genetic diversity due to genetic drift. In addition to population size, a lack of gene flow between populations may be responsible for the observed levels of heterozygosity. Barriers to migration can strongly influence levels of variability within bull trout populations (Costello et al. 2003; Neraas and Spruell 2001; Meeuwig et al. 2010) and have been shown to explain higher levels of genetic

diversity than other landscape features (e.g., stream order, watershed area, stream temperature, road density; Costello et al. 2003). Currently, bull trout populations in Pine Creek, Indian Creek, and Wildhorse River basins exhibit a resident life history (Buchanan et al. 1997; Chandler et al. 2001, 2003; Wilkison and Trainer 2017; IPC 2020), which in conjunction with small population sizes, may explain the lower levels of observed heterozygosity.

Similar processes likely drove the patterns of genetic differentiation observed in our study to those that shaped levels of genetic diversity. Bull trout generally display high levels of genetic differentiation among populations, and differentiation can occur at small spatial scales (e.g., Neraas and Spruell 2001). Patterns of genetic differentiation in bull trout populations are expected to be driven by a combination of environmental factors (e.g., migration barriers, waterway distance between populations, and patch size), demographic parameters (e.g., population size, expression of multiple life histories), and long-term evolutionary patterns (e.g., glacial history) (Neraas and Spruell 2001; Costello et al. 2003; Whiteley et al. 2006; Meeuwig et al. 2010; Ardren et al. 2011; Kovach et al. 2015). In our system, we observed a wide range of genetic differentiation values at both the reporting unit and population levels (Figs. 3, 4). Three of our study's four most highly differentiated populations were identified as having minimal available spawning habitat (Indiana and Limber Jim creeks) or consistently low densities (Hurricane Creek; ODFW 2005). In contrast, the Imnaha River population had the lowest levels of genetic differentiation and supports one of Oregon's most robust bull trout populations (ODFW 2005). The ability to compare levels of F_{ST} from this study to others is limited by the fact that few studies have employed the same GT-seq panel; however, when this marker panel was developed, values of F_{ST} generated using this marker set were larger than, but highly correlated with, estimates generated using microsatellites ($r = 0.896$, Bohling et al. 2021). The SNPs panel used here was designed for monitoring neutral genetic variation in bull trout populations (e.g., estimating levels of genetic diversity, differentiation, and genetic assignment tests; Bohling et al. 2021). Research in other salmonids from the Pacific Northwest has identified regions of the genome associated with adaptive traits (e.g., run timing in steelhead *Oncorhynchus mykiss* and Chinook Salmon *O. tshawytscha*; Narum et al. 2024); however, similar efforts to identify loci under selection in bull trout have lagged. Moving forward, additional analyses are necessary to understand what combination of demographic (e.g., population size) or environmental factors (e.g., geographic distance) best explain observed patterns of differentiation examined in our study system and describe adaptive variation across the landscape.

Owing to the high levels of genetic differentiation commonly observed among populations of bull trout, genetic stock identification, and mixture analysis have proven both effective and accurate at identifying source-of-origin for fisheries mixtures (DeHaan et al. 2011a, b; Warnock et al. 2011; Nyce et al. 2013; Bohling et al. 2021; Taylor et al. 2021). In our study, self-assignment tests placed fish into their known reporting unit of origin for all but three (> 99.8%) of the samples in our baseline, and assignment accuracy at the population level was also relatively high (96.5%). The few misassigned fish at the reporting unit level occurred between neighboring drainages (Grande Ronde and Imnaha rivers), consistent with low pairwise genetic differentiation levels between these two basins. The high self-assignments rates and the tendency for misassigned fish to occur in proximate basins mirror results from other river systems. For example, ~ 90% of bull trout were assigned to the correct tributary of origin, and ~ 97% of fish were assigned within the correct management region in the Clark Fork River system (DeHaan et al. 2011a). Relative to previous microsatellite baselines, the GT-seq panel used here has performed equivalently well or better based on surveys from the lower/upper portions of the Clark Fork River as well as the Skagit River (Bohling et al. 2021). These results suggest that mixture analysis of fish captured below dams can be used to inform upstream passage efforts with limited concern for misassignment. Upstream passage programs for bull trout that use real-time genetic analysis to identify migrants from above-dam tributaries are currently few (Lewis River, Washington; DeHaan and Adams 2011; Adams and DeHaan 2015; and Clark Fork River, Idaho/Montana; Adams et al. 2021); however, fish passed upstream have been shown to make significant reproductive contribution within natal streams (DeHaan and Bernall 2013). While there is currently no upstream passage at Hells Canyon Dam, we identified the Imnaha River as a primary source of migratory bull trout sampled from the mainstem Snake River, which is promising in that the Imnaha River Species Management Unit is one of the most abundant and viable in Oregon (ODFW 2005).

Taken together, results from this study suggest that upstream passage of migratory bull trout from below Hells Canyon Dam could promote gene flow, which can encourage the expression of multiple life histories and increase genetic diversity and adaptive potential in recipient populations. While upstream passage resulting in gene flow is straightforward in theory, the specifics of how and where fish are released and the methods used to quantify population-level impacts warrant consideration. Genetic rescue, a term used to describe the reduction in extinction probability of a population as a result of gene flow (Whiteley et al. 2015; Bell et al. 2019), has been documented in multiple taxa (Whiteley et al. 2015), including fish species (Fitzpatrick et al. 2016; Robinson et al. 2017). Recent work has identified criteria to select populations that would most benefit from genetic rescue (Kovach et al. 2022) and appropriate metrics to quantify

near- and long-term success (Robinson et al. 2021). For Pine Creek, Indian Creek, and Wildhorse River basins, migratory bull trout downstream of Hells Canyon Dam could be passed above the dam and allowed to spawn volitionally in headwater streams. Alternatively, downstream bull trout could be transported into pre-selected streams based on genetic data presented here (i.e., populations with low expected heterozygosity and number of polymorphic markers; Kovach et al. 2022) with untreated streams serving as experimental controls to account for environmental variation (e.g., Robinson et al. 2017). Ultimately, resource availability and permitting will likely determine the specific approach implemented. Still, the upstream passage of migratory bull trout above Hells Canyon Dam represents an opportunity to add to the growing body of literature on using genetic data to conserve imperiled populations.

Conclusions

The most recent Recovery Plan for the coterminous United States bull trout population identified historical habitat loss and fragmentation, interaction with nonnative species, and fish passage issues as the most significant primary threats affecting bull trout (USFWS 2015). Upstream passage of migratory adults has the potential to maintain distinct life histories within bull trout populations and minimize the impacts of migration barriers. Still, there remains a paucity of empirical studies on the subject. Importantly, where implemented, upstream passage has been shown to increase spawning adults in small, isolated populations above dams, significantly contributing to reproduction (DeHaan et al. 2011a; DeHaan and Bernall 2013). Our study objectives were to generate a genetic baseline to inform passage efforts for migratory bull trout in the Hells Canyon Reach of the Snake River and to detail nearby populations' genetic structure and diversity. We observed lower levels of genetic diversity in populations above Hells Canyon Dam relative to downstream basins, which is not uncommon among isolated headwater populations of bull trout (e.g., Neraas and Spruell 2001). However, we also identified populations in the Pine Creek and Wildhorse River basins with observed heterozygosity roughly equal to or greater than the baseline average, suggesting that reservoirs of diversity remain in headwater drainages. Genetic stock identification performed on migratory bull trout sampled from below Hells Canyon dam assigned these fish almost exclusively (> 97%) to the Imnaha River, a finding significant for several reasons. First, the Imnaha River boasts one of Oregon's most robust bull trout populations (ODFW 2005) and contains the highest levels of genetic diversity observed in this study. Second, bull trout above Hells Canyon Dam were genetically more similar to bull trout in the Imnaha River than other downstream populations. These findings imply that the Imnaha River could be an appropriate donor population for upstream passage (i.e., demographically stable and genetically robust) that could increase genetic diversity and adaptive potential if individuals reproduce in headwater populations above Hells Canyon Dam. Moving forward, the genetic resources described here detail the genetic structure of bull trout populations above and below the Hells Canyon Dam, which can be used to inform ongoing restoration and conservation efforts, including upstream passage, translocation, and in-stream barrier removal projects.

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Author contribution All authors contributed to the design and execution of the study. Data analysis was performed by Thomas Delomas, John Hargrove, Audrey Harris and Matthew Campbell. The first draft of the manuscript was written by John Hargrove and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Data availability The genetic baseline is freely available on fishgen.net. Users must log into the website, and the data set can be found under 'Data Sets' by searching for 'IPC Hells Canyon Bull Trout GSI Baseline v1.0 rev 5/2021' (Dataset Id: 20210316).

Declarations

Conflict of interest The authors declare no conflict interest.

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